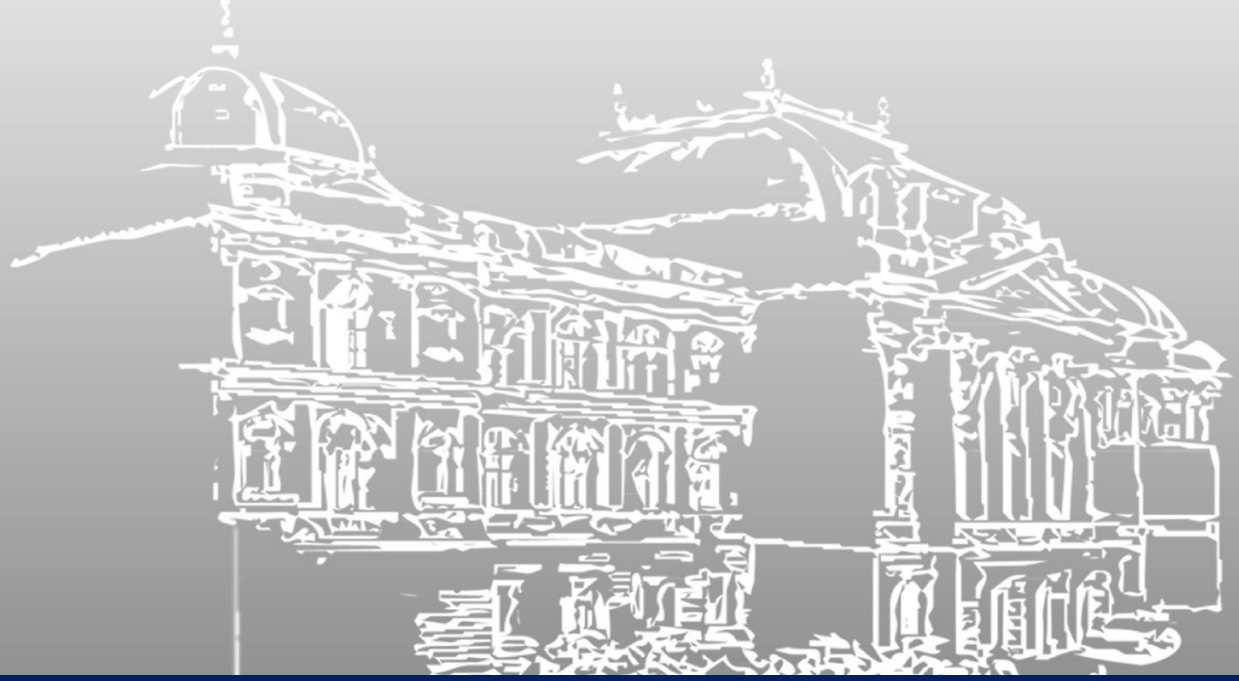


Consumer Decision Making

Elisabeth Lex

AI for Society Lab

https://youtu.be/cHiWsCkBWi8

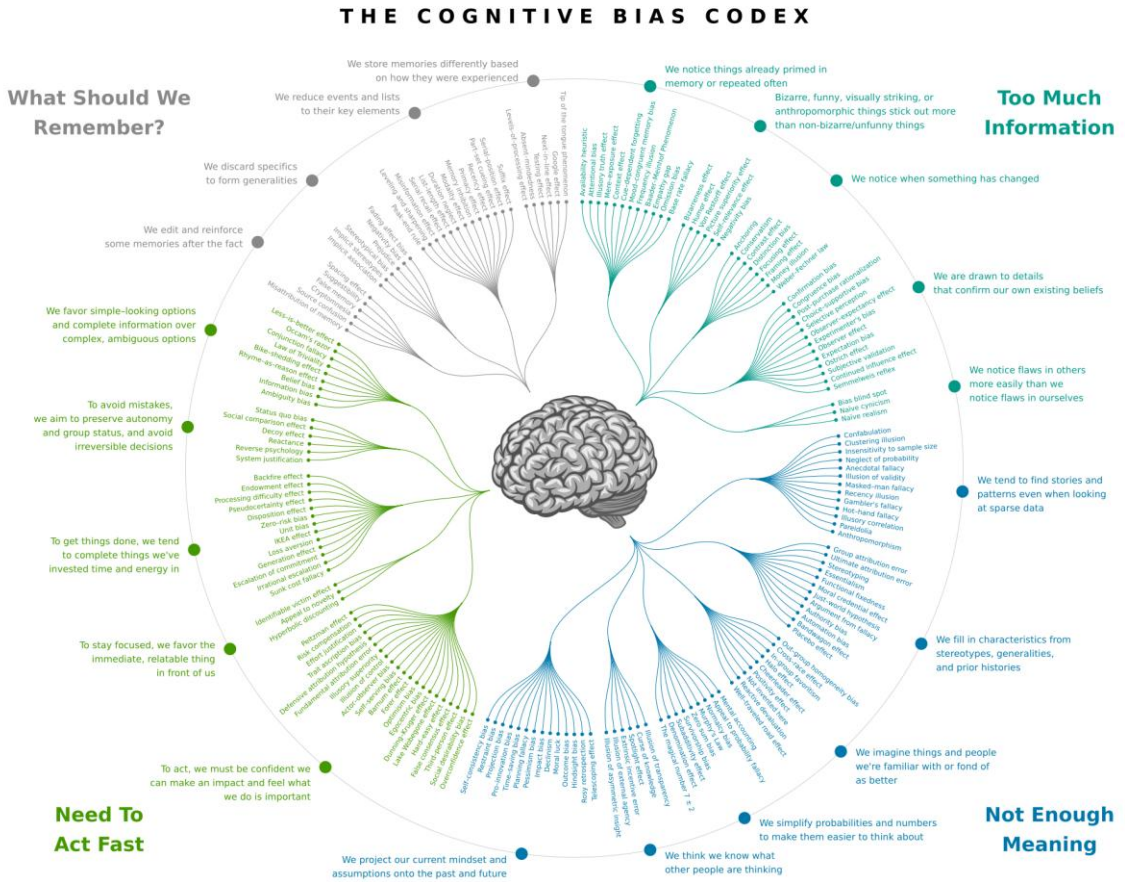
Agenda

1. Basic introduction to cognitive biases
2. Case studies of such biases in recommender systems

Cognitive Biases

- Tendency to decide in certain (simplified) ways
- Good overview of biases that influence our decision making:

https://yourbias.is



Why can biases be problematic?

"… people rely on a limited number of heuristic principles which reduce the complex tasks of assessing probabilities and predicting values to simpler judgmental operations. In general, these heuristics are quite useful, but sometimes they lead to severe and systematic errors."

Tversky & Kahneman (1974, p. 1124)

Why Cognitive Biases?

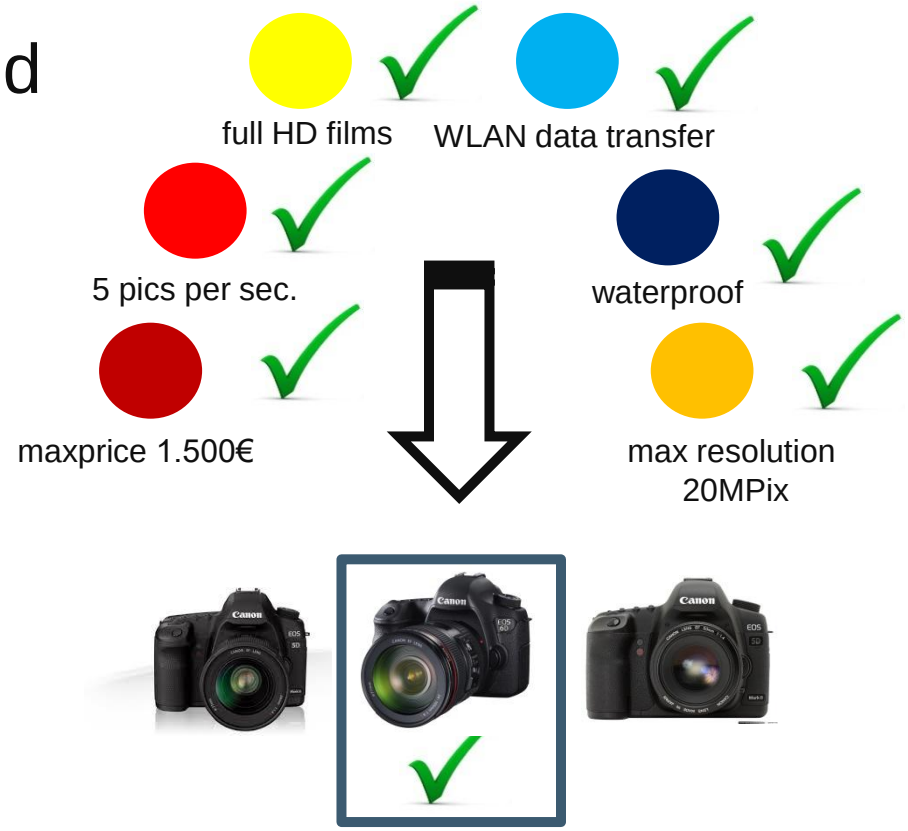


Human brains were not primarily designed for the present time but rather for stone-age conditions

Frequent Assumptions …

- Preferences are known/defined beforehand
- Preferences are stable, users don't change them
- Users have an optimization function in mind

☞ **However, preference stability does not exist!**



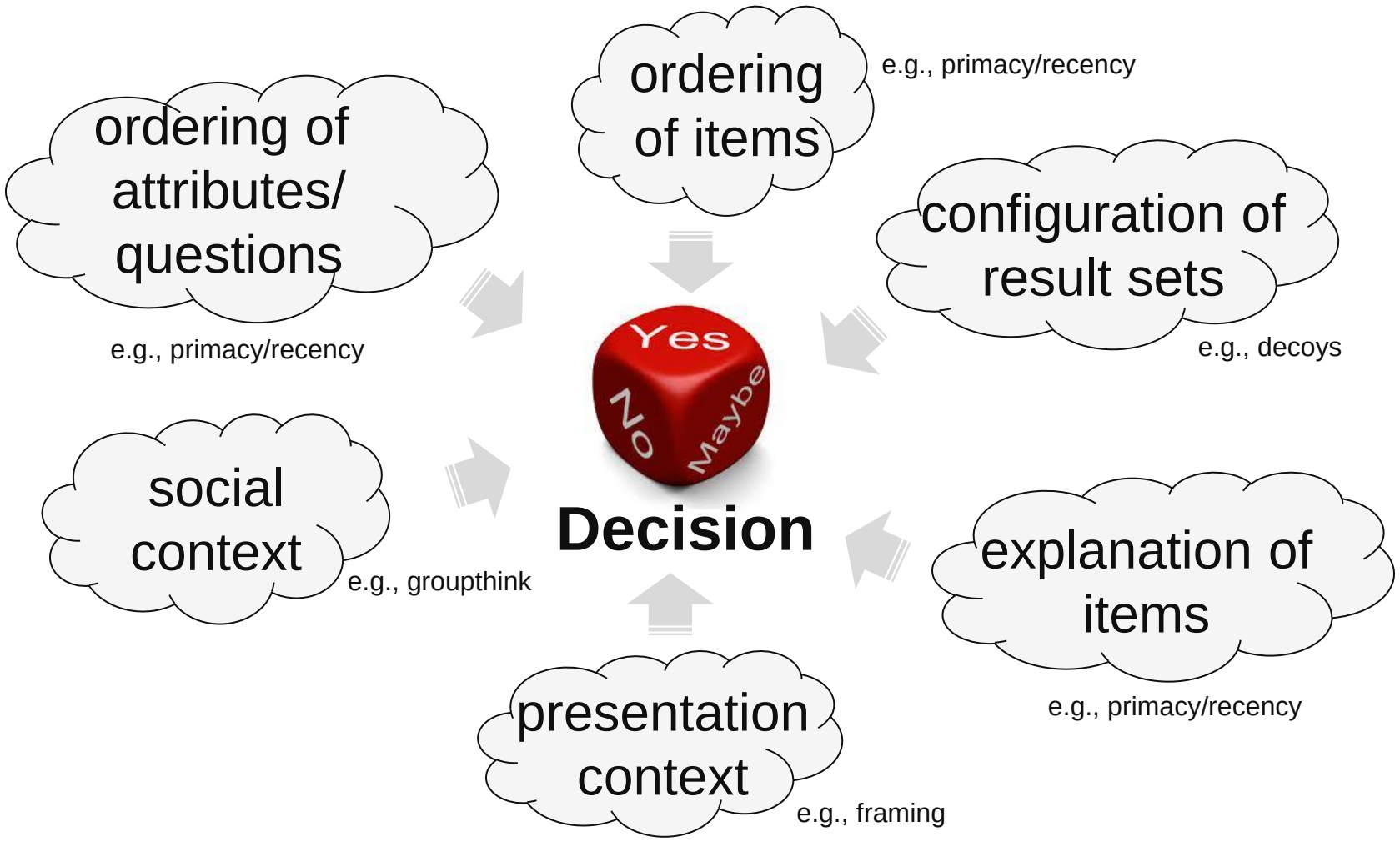
Preferences Are Constructed …

- Not known beforehand
- Often changed
- No optimization function used
- Decision heuristics applied (e.g., elimination by aspects, attribute-wise elimination of items, e.g., price < 300)

☞"Door opener" for **cognitive biases** (tendency to decide in certain ways)!

J. Payne, J. Bettman, and E. Johnson. The Adaptive Decision Maker, Cambridge University Press, 1993.

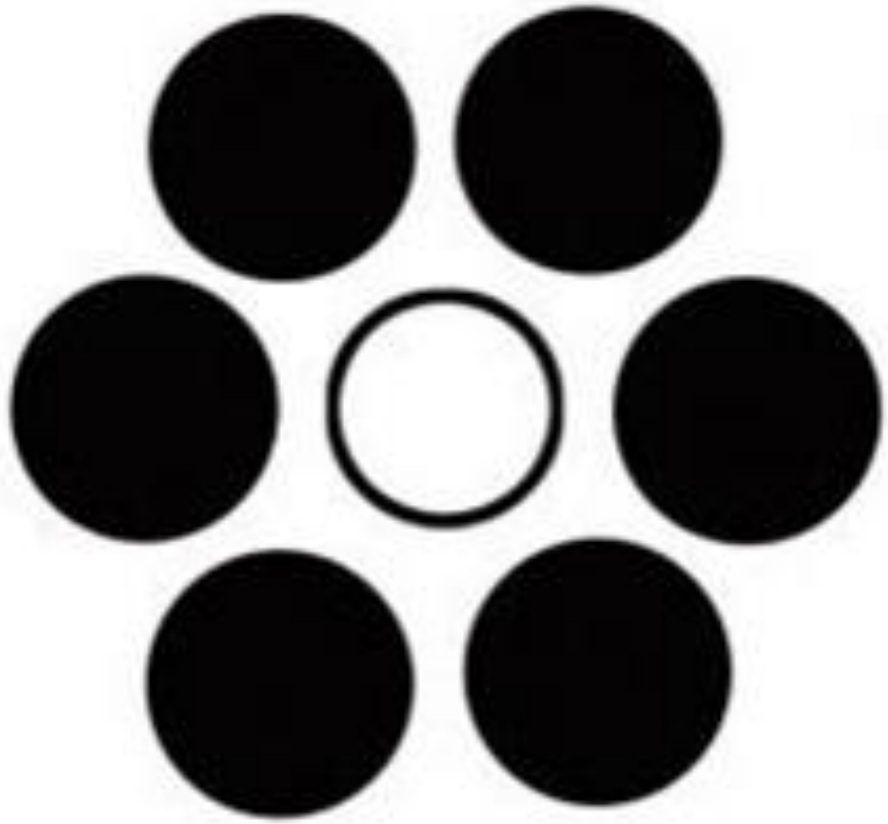
Influence Factors for Decisions with Recommender Systems



Examples of Cognitive Biases

Theory	Description
Context effects (decoy effects)	Additional irrelevant (inferior) items in an item set significantly influence the selection behavior
Primacy/recency effects	Items at the beginning and the end of a list are analyzed significantly more often than items in the middle of a list
Framing effects	The way in which different decision alternatives are presented influences the final decision taken
Priming	If specific decision properties are made more available in memory, this influences a consumer's item evaluations
Defaults	Preset options bias the decision process

Context Effects



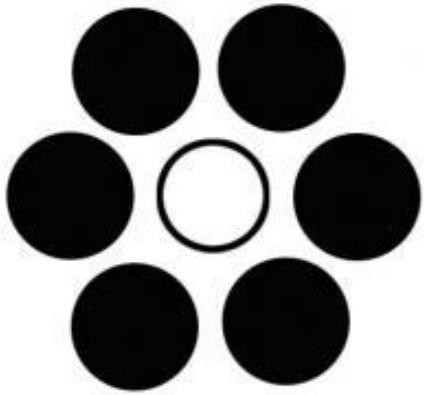
Context Effects

- A decision is always made depending on the context in which item alternatives are presented
- For example, completely inferior item alternatives can trigger significant changes in choice behaviors
- Examples of context effects are discussed in the following

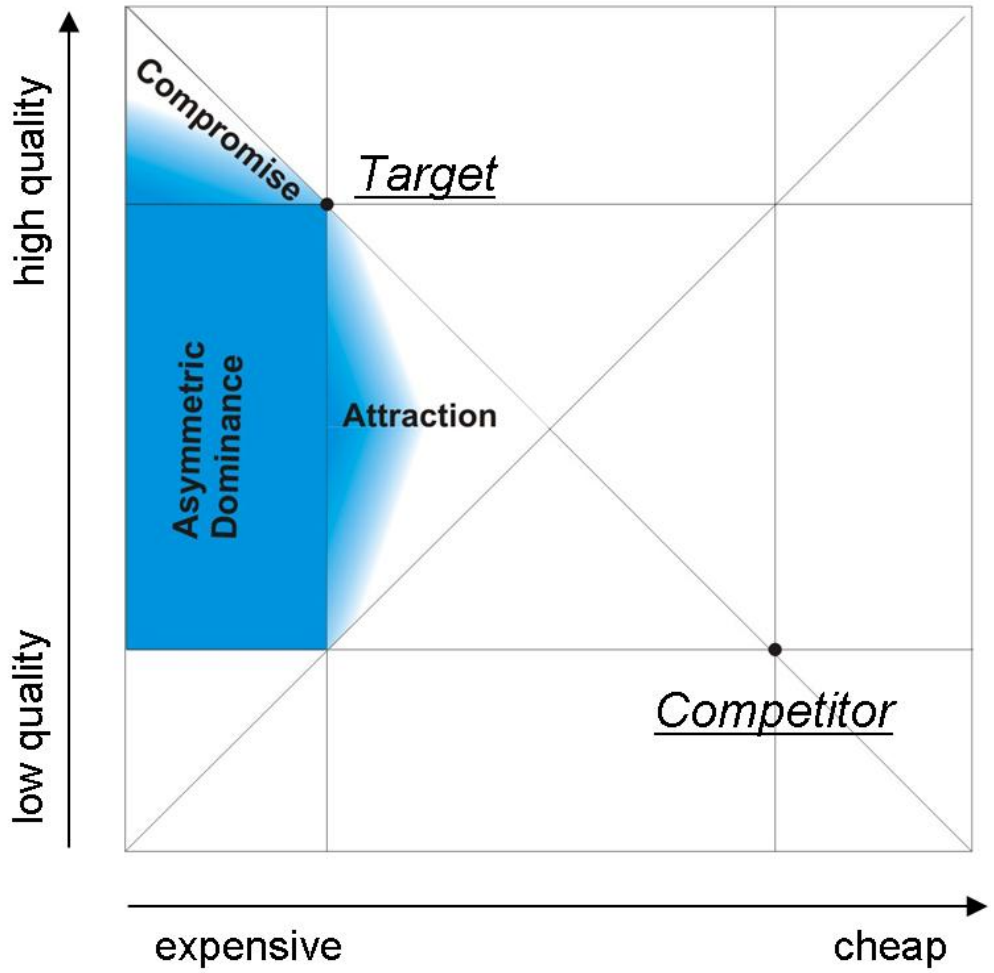
Short Note: Ebbinghaus Effect

- Illusion of relative size perception
- Triggered by context in which objects are shown
- Commonalities with context effects





Context Effects: Overview



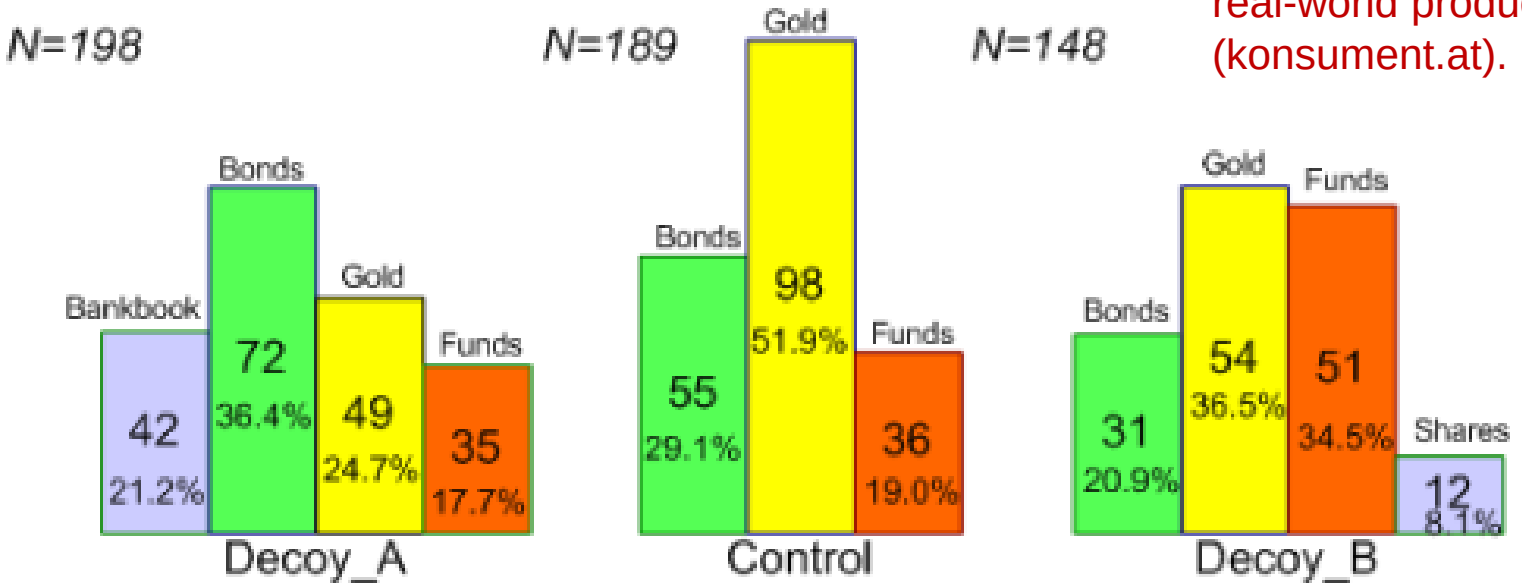
- **Compromise** : Target (T) is a compromise to decoy item D (T is **less expensive** and has **slightly lower quality**)
- **Asymmetric Dominance**: T dominates D (T is **cheaper** and has a **higher quality**)
- **Attraction**:
  T is more attractive than D (T is **slightly more expensive** but has a **higher quality**)

Compromise Effect

Product	A (T)	B	D
price per month	30	15	50
download limit	10GB	5GB	12GB

- The addition of alternative *D* (the decoy alternative) increases the attractiveness of alternative *A* because, compared with product *D*, *A* has only a slightly lower download limit but a significantly lower price
- *D* is a so-called decoy product, which represents a solution alternative with the lowest attractiveness

Compromise Effect in the Financial Services Domain



Study performed with real-world products (konsument.at).

A. Felfernig, E. Teppan, and K. Isak. Decoy Effects in Financial Service e-Sales Systems, ACM Recommender Systems Workshop on Human Decision Making and Recommender Systems (Decisions@RecSys), Chicago, IL, 2011.

Asymmetric Dominance Effect

Product	A (T)	B	D
price per month	30	15	50
download limit	10GB	5GB	9GB

- Product *A* dominates *D* in both dimensions (price and download limit)
- Product *B* dominates alternative *D* in only one dimension (price)
- The additional inclusion of *D* into the choice set could trigger an increase of the selection probability of *A*

Asymmetric Dominance Effect



Attraction Effect

Product	A (T)	B	D
price per month	30	90	28
download limit	10GB	30GB	7GB

- Product *A* is a little bit more expensive but of significantly higher quality than *D*
- Introduction of product *D* would induce an increased selection probability for *A*

Impacts on Recommender Systems

- Faster decisions: decoys help to resolve cognitive dilemmas in the case of items with the same utility
- Increased confidence: decoys serve as a basis for explaining a decision
- Increased share of specific items: systematic “push” of target items
- Diagnosis support: figuring out which items are responsible for the low share of a target item
- Interferences between different decoy items in a set

A. Felfernig, B. Gula, G. Leitner, M. Maier, R. Melcher, S. Schippel, E. Teppan. A Dominance Model for the Calculation of Decoy Products in Recommendation Environments. AISB Symposium on Persuasive Technologies, Vol. 3, pp. 43-50, Aberdeen, Scotland, Apr. 1-4, 2008.

Primacy/Recency Effects

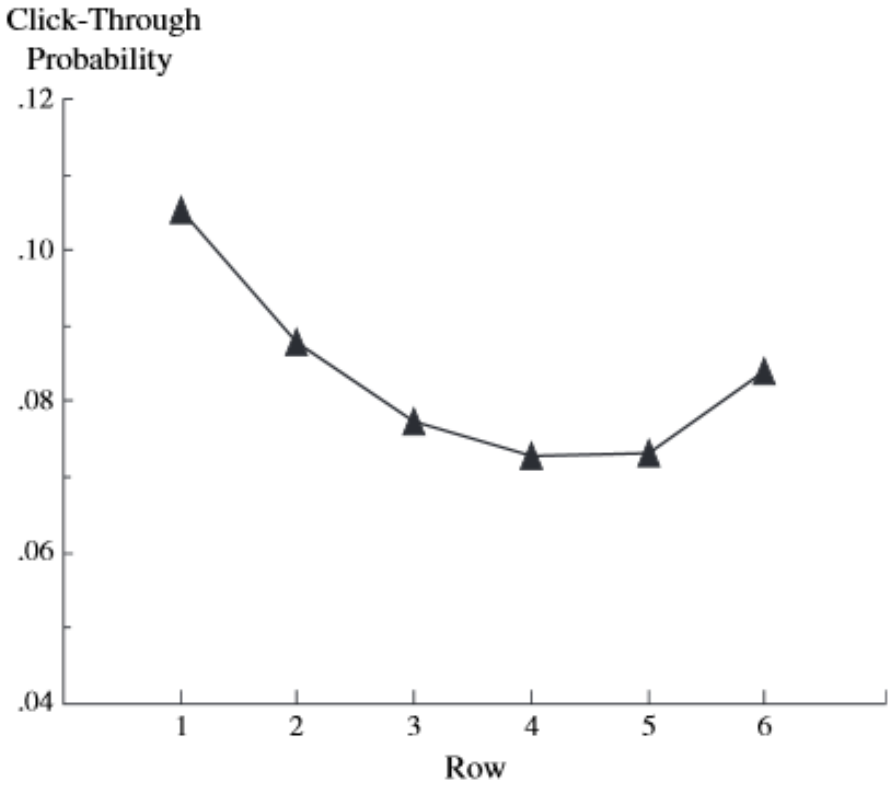


Primacy/Recency Effects as a Decision Phenomenon

- Describe situations in which items presented at the beginning and at the end of a list are evaluated significantly more often than others
- Typically, users are not interested in evaluating large lists to identify those that best fit their preferences
- The same phenomenon exists in many other contexts as well:
  - Web search
  - Marketing
  - …

Item Selection Behavior (Web Links)

- Primacy effect
- Efficacy of the first link
- But also recency
- Tendency to click links at the end



J. Murphy, C. Hofacker, and R. Mizerski. Primacy and Recency Effects on Clicking Behavior. Computer-Mediated Communication, 11:522-535, 2012.

Primacy/Recency Effects as a Cognitive Phenomenon

- Describe situations in which information units at the beginning (primacy) and at the end (recency) of a list are recalled more often than information units in the middle of the list
- Primacy/recency effects in recommendation dialogs must be taken into account because different dialog sequences can potentially change the selection behavior of consumers

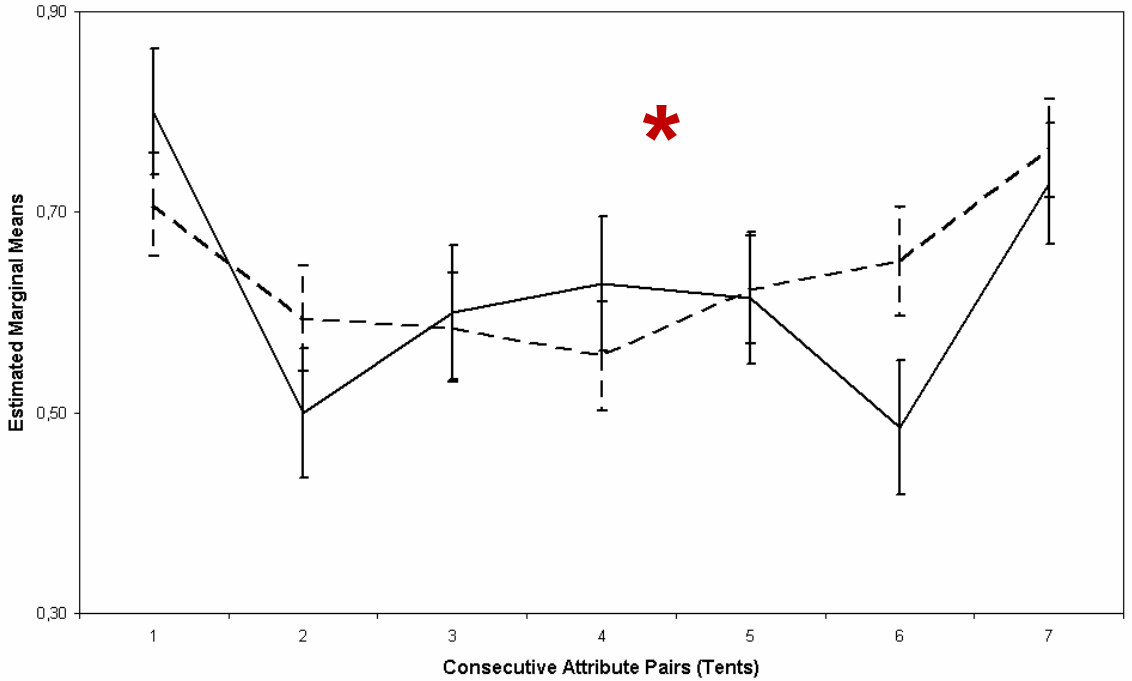
A. Felfernig, G. Friedrich, B. Gula, M. Hitz, T. Kruggel, R. Melcher, D. Riepan, S. Strauss, E. Teppan, and O. Vitouch. Persuasive Recommendation: Exploring Serial Position Effects in Knowledge-based Recommender Systems, Second International Conference of Persuasive Technology (Persuasive 2007), Springer Lecture Notes in Computer Science, Vol. 4744, pp.283-294, Stanford, California, Apr. 26-27, 2007.

Primacy/Recency Effects as a Cognitive Phenomenon

- Descriptions at beginning/end of dialog are recalled more often

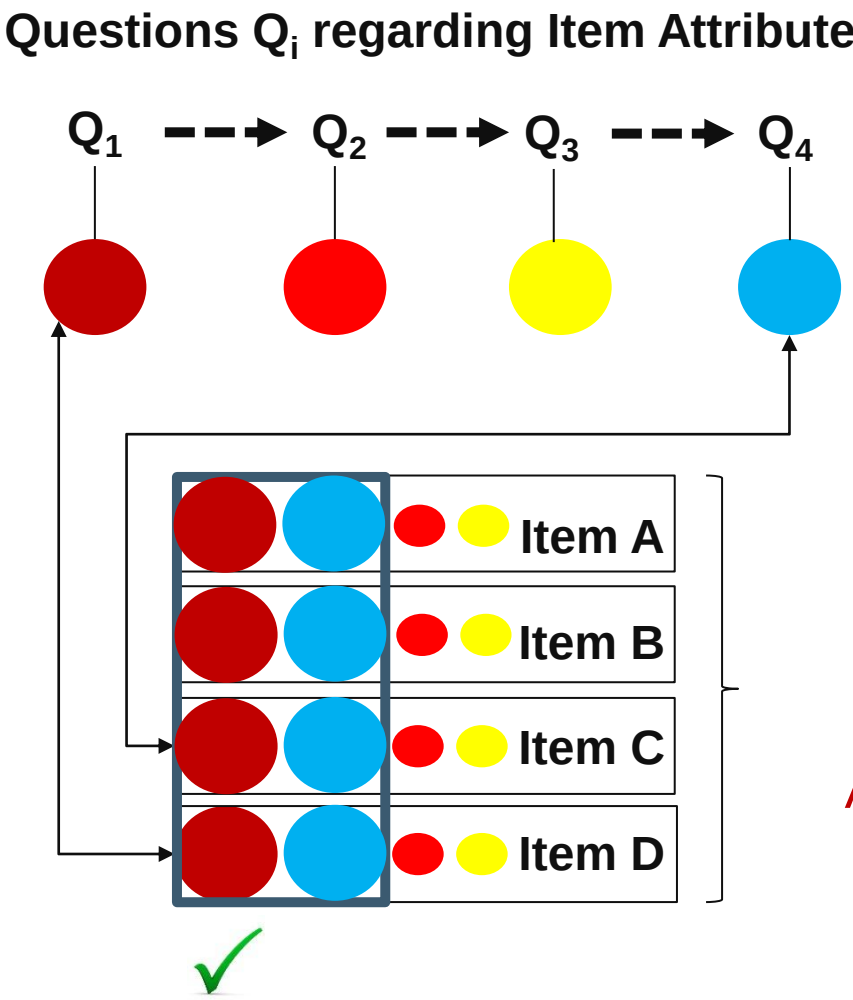


- Also in the case “unfamiliar salient” (*), e.g. flyscreen vs. price or weight.



A. Felfernig, G. Friedrich, B. Gula, M. Hitz, T. Kruggel, R. Melcher, D. Riepan, S. Strauss, E. Teppan, and O. Vitouch. Persuasive Recommendation: Exploring Serial Position Effects in Knowledge-based Recommender Systems, Second International Conference of Persuasive Technology (Persuasive 2007), Springer Lecture Notes in Computer Science, Vol. 4744, pp.283-294, Stanford, California, Apr. 26-27, 2007.

Impacts on Item Selection



A. Felfernig, G. Friedrich, B. Gula, M. Hitz, T. Kruggel, R. Melcher, D. Riepan, S. Strauss, E. Teppan, and O. Vitouch. Persuasive Recommendation: Exploring Serial Position Effects in Knowledge-based Recommender Systems, 2nd International Conference of Persuasive Technology (Persuasive 2007), Springer Lecture Notes in Computer Science, Vol. 4744, pp.283-294, Stanford, California, Apr. 26-27, 2007.

Attribute order has an impact on perceived attribute importance (e.g., price, weight, …)!

Example: Restaurant A

Breathtaking view over the city center. Easily accessible via the highway, plenty of parking opportunities. Terrific ambiente with an open-view kitchen. Side dishes and sauces can be freely customized. Friendly service. Upper price segment. Food quality does not justify the price. Personel leaves the impression of unkemptness. Poor cleanliness. Elegant furniture and a good place for relaxing.

Example: Restaurant B

Upper price segment. Food quality does not justify the price. Breathtaking view over the city center. Easily accessible via the highway, plenty of parking opportunities. Terrific ambiente with an open-view kitchen. Side dishes and sauces can be freely customized. Friendly service. Elegant furniture and a good place for relaxing. Personel leaves the impression of unkemptness. Poor cleanliness.

M. Stettinger, A. Felfernig, G. Leitner, S. Reiterer, and M. Jeran. Counteracting Serial Position Effects in the CHOICLA Group Decision Support Environment, 20th ACM Conference on Intelligent User Interfaces (IUI2015), pp. 148-157, Atlanta, Georgia, USA, 2015.

Impacts on Recommender Systems

- Control of item selections on the basis of attribute orderings in dialogs
- Control of diagnosis & repair and critique selection
- Users rate items differently depending on the ordering of argumentations in reviews

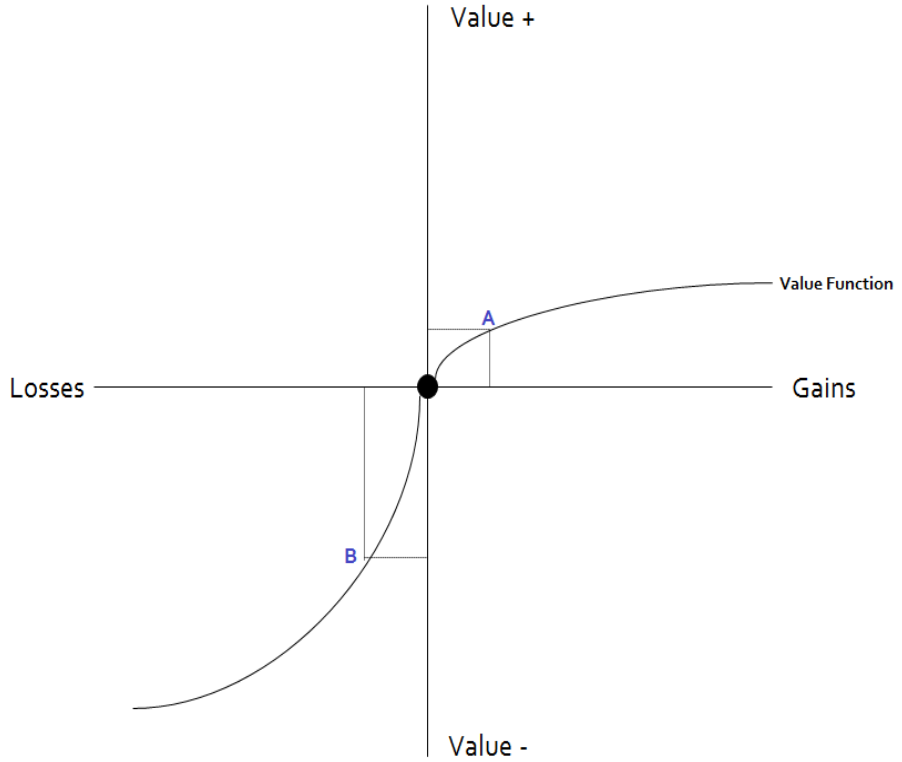
A. Felfernig, G. Friedrich, B. Gula, M. Hitz, T. Kruggel, R. Melcher, D. Riepan, S. Strauss, E. Teppan, and O. Vitouch. Persuasive Recommendation: Exploring Serial Position Effects in Knowledge-based Recommender Systems, 2nd International Conference of Persuasive Technology (Persuasive 2007), Springer Lecture Notes in Computer Science, Vol. 4744, pp.283-294, Stanford, California, Apr. 26-27, 2007.

Framing



Framing

- Framing Effect: the way a decision alternative is presented influences the decision behavior of the user
- Example: 80% lean vs. 20% fat meat
- Prospect theory: suggests that potential purchases are evaluated in terms of **gains** or **losses** (see “price framing” …)



D. Kahneman und A. Tversky (1979): Prospect theory: An analysis of decision under risk, Econometrica, Vol. 47, No. 2, S. 263-291.

Price Framing: Example

•Which company would you purchase wood pellets from, X or Y?

- Company X sells pellets for **€24.50** per 100kg, and gives a **€2.50 discount** if the customer pays with cash
- Company Y sells pellets for **€22.00** per 100kg, and charges a **€2.50 surcharge** if the customer uses a credit card

☞Company **X rewards buyers** with a discount, which is considered a gain (we want to avoid losses …)

M. Bertini and L. Wathieu. The Framing Effect of Price Format. Working Paper, Harvard Business School, 2006.

Impacts on Recommender Systems

- Positive framing increases selection probability (e.g., 95% no loss vs. 5% loss) → use graphical representation …

- Price framing: potential shift from quality to secondary attributes (e.g., payment services)

Priming


Virtual ShowPlace Stores - Netscape
File Edit View Go Communicator Help
Back Forward Reload Home Search Netscape Print Security Stop
Instant Message Marketing 101 H Naomi's Page Penn Library-Ho University of P Philadelphia On Excite Search E
Virtual ShowPlace Stores
A Whole New Way To Buy Furniture
Virtual ShowPlace Stores and In-Store Galleries
From the moment you walk into a Virtual ShowPlace Store, or a Virtual ShowPlace Gallery inside a participating furniture store, you'll notice one thing. Here's a place where it's easy to find a sofa you'll like at a price you can afford. We say it's "a whole new way to buy furniture."
You'll start by choosing the sofa style that's right for you. Overstuffed or casual? Formal or tailored? Transitional in an eclectic setting? You may even find just the right style in just the right fabric. Wow, it is that easy! But maybe you're not too crazy about the fabric the store put on your sofa. Now the fun begins. Virtual has over 500 designer fabrics. No need to limit your imagination. And no need to break the bank - our sofas generally retail from $600 to $1200 - hundreds less than you might expect to pay for the style, comfort and quality you'll get with Virtual.
Click here to see our products
Document: Done

Priming

- Idea of making some properties of a decision alternative more accessible in memory such that this setting will directly influence user evaluations
- Def. Influencing of the processing of a current stimulus by the activation of already memorized knowledge by a precedent stimulus
- Example: background priming exploits the fact that different page backgrounds can directly influence the decision-making process

Background Priming

- ☞ Cloudy background triggered user feelings of comfort and caused users to select more expensive products (focus on quality attributes)





N. Mandel and E. Johnson. Constructing Preferences online: Can Web Pages Change What You Want? Association for Consumer Research Conference, Montreal, pp. 1-37, 1998.

A. North, D. Hargreaves, and J. McKendrick. In-store music affects product choice. Nature 390:132, 1997.

More Effects

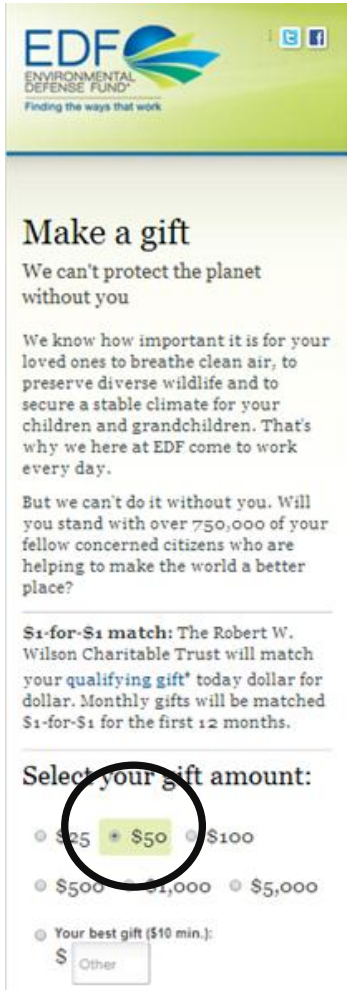


"All those in favor say 'Aye.'"
"Aye."
"Aye."
"Aye."
"Aye."
"Aye."

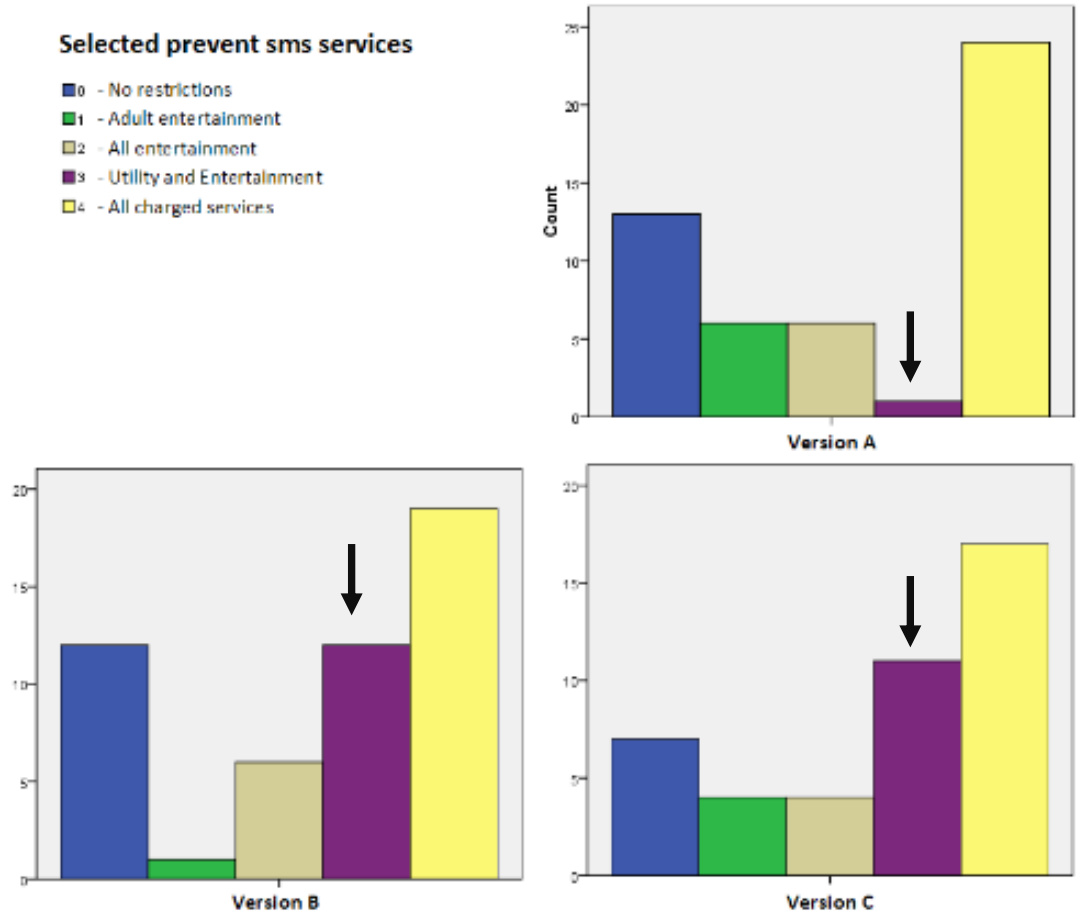
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Defaults

- People tend to favor the status quo compared to other decision alternatives (“status quo bias”)
- People are typically loss-averse (prospect theory)
- If defaults are used, users are reluctant to change predefined settings (mistakes, additional effort, …)
- Defaults can be used, for example, to …
  - Influence decisions (ethical issues!)
  - Reduce the overall interaction effort and actively support consumers in the product selection process



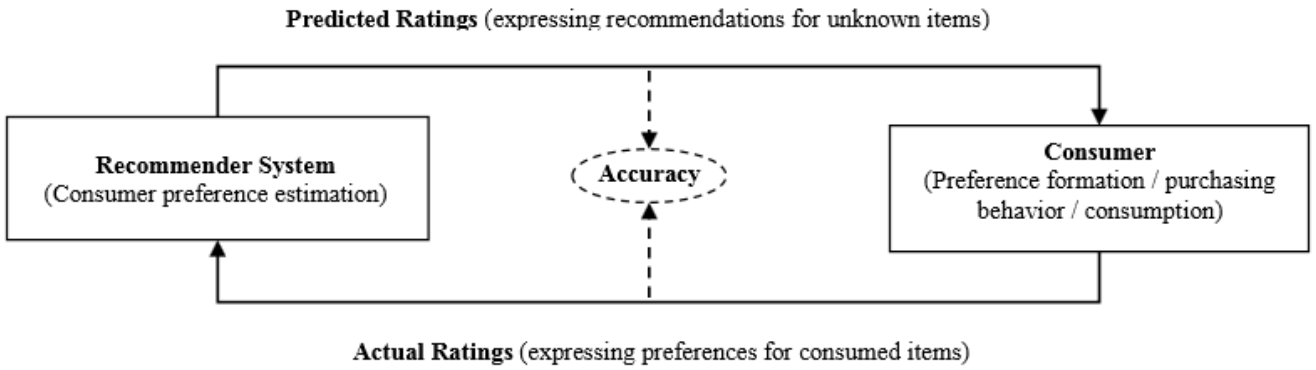
Defaults: Example



M. Mandl, A. Felfernig, and J. Tiihonen: Evaluating Design Alternatives for Feature Recommendations in Configuration Systems. CEC 2011, pp. 34-41, 2011.

Anchoring

- Tendency to rely too heavily on the first information (anchor) within the scope of decision making
- Ratings biased to be higher result in higher ratings of the current user
- Example: ratings in collaborative filtering, preferences articulated by the first group member



G. Adomavicius, J. Bockstedt, S. Curley, and J. Zhang. Recommender Systems, Consumer Preferences, and Anchoring Effects, Decisions@RecSys'11, pp. 35-42, Chicago, IL, USA, 2011.

A. Felfernig, C. Zehentner, G. Ninaus, H. Grabner, W. Maaleij, D. Pagano, L. Weninger, and F. Reinfrank, Group Decision Support for Requirements Negotiation, LNCS, 7138, pp.105-116, 2012.

GroupThink

→ The desire for group harmony results in a suboptimal decision. Group members try to minimize conflict and take a decision without an in-depth and critical evaluation of the alternatives.



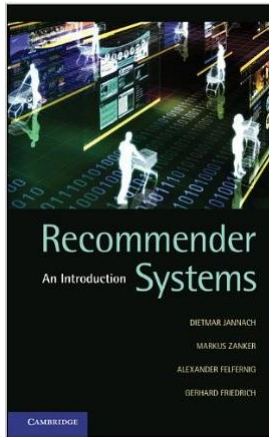
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Literature

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- [Chen and Pu 2005] Trust building in recommender agents, 1st International Workshop onWeb Personalisation, Recommender Systems and Intelligent User Interfaces (WPRSIUI ’05) (Reading, UK), 2005, pp. 135–145
- [Chen and Pu 2008] A cross-cultural user evaluation of product recommender interfaces, Proceedings of the 2008 ACM Conference on Recommender Systems (RecSys ’08) (Lausanne, Switzerland), ACM, 2008, pp. 75–82
- [Cosley et al. 2003] Is seeing believing? How recommender system interfaces affect users’ opinions, Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI ’03) (Fort Lauderdale, FL), 2003, pp. 585–592
- [Felfernig et al. 2008] Persuasion in knowledge-based recommendation, Proceedings of the 2nd International Conference on Persuasive Technologies (Persuasive ’08) (Oulu, Finland), vol. 5033, Springer, 2008, pp. 71–82
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- [Payne et al. 1993] The adaptive decision maker, Cambridge University Press, 1993
- [Simon 1955] A behavioral model of choice, Quarterly Journal of Economics 69 (1955), no. 1, 99–118

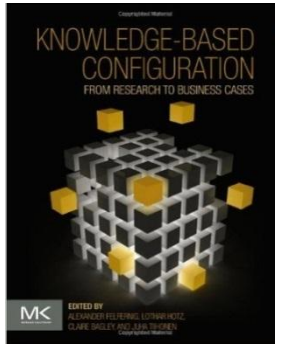
Course Material

- D. Jannach, M. Zanker, A. Felfernig, and G. Friedrich. Recommender Systems – An Introduction, Cambridge University Press, 2010 (*can be found in library*).



- Lecture slides and slides from recommenderbook.net.

- A. Felfernig, L. Hotz, C. Bagley, and J. Tiihonen. Knowledge-based Configuration – From Research to Business Cases, Elsevier/Morgan Kaufmann, 2014.



Exercise 1

- Select an example domain for a knowledge-based recommender application.
- Define two products (items): a target product T and a competitor product C.
- Define the following decoy item:
  - A decoy item that triggers asymmetric dominance

- Note: no submission („Abgabe“) needed!

Exercise 2

- Explain the term “decision bias” on the basis of an example.
- Explain and provide an example of: primacy/recency, anchoring, and defaults.

- Note: no submission („Abgabe“) needed!

Obligatory Exercises

•Each student should participate in the online exercises on „Consumer Decision Making“. The points from this challenge are part of your evaluation (Übungsteil).

- https://checkr.tugraz.at/
- Participation key: ~~**H9aK71**~~ Correct Key is dp2u3q
- **(TU Graz single sign on)**

Thank you for your attention!